INDEX NUMBER

LANKA NIPPON BIZTECH INSTITUTE

FINAL EXAMINATION

Department	Computing
Course	BSc (Hons) Software Engineering
Module Code	FND 210130
Module Name	Mathematics III for SE
Year of Study	2
Academic Year	2024
Semester	1
Date of the Examination	3rd of September, 2024
Time of the Examination	9.00 am – 12.00 pm
Duration	**3 hours**
Maximum marks for final grade	60

Instructions to Candidates

1. This paper contains 04 questions on 02 pages (from page 2 to 3).
2. You should **answer all the questions.**
3. Student Should write answers according to the order of the question paper.
4. Clearly write your index number (do not write your name or student ID) on the examination booklet, supplementary answer sheets and the question paper.
5. Write your answers in the examination answer booklet provided and all answers to a new question should start on a new page.
6. Clearly write the question number on the left side margin of the answer sheet.
7. Candidates are not allowed to bring any material other than those allowed by the invigilator into the examination hall.
8. Students are not permitted to use red ink / correction pens while writing answers.
9. This is a closed book examination.
10. Please check you have entered all the necessary information on your answer booklet and attached all the supplementary answer sheets.
11. You are not allowed to take the examination paper or any other material out of the examination hall.

Question 1

a) Let A and B are matrix such that , $A = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$, $B = \begin{bmatrix} x & 1 \\ 4 & -1 \end{bmatrix}$ and if $(A + B)^2 = A^2 + B^2$ is given, then find the value of x.

[6 marks]

b) Let A and B are two 3×3 matrices which satisfied $3A = A^2 + AB$. Further it is given that A is nonsingular matrix and $B = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 2 & -1 \\ -1 & 0 & 1 \end{bmatrix}$.
Find the matrix A and the inverse of A.

[10 marks]

c) Solve the following system of equations using **Gaussian Elimination Method.**

$$\begin{aligned} 3x - 2y &= 10 \\ 2x + y &= 2 \end{aligned}$$

[6 marks]

d) Solve the following system of equations using **Cramer's Rule.**

$$\begin{aligned} x + y + z &= 6 \\ 2x - y + z &= 3 \\ x + 2y - z &= 3 \end{aligned}$$

[8 marks]

[Total – 30 marks]

Question 2

a) For given two points $A = (-4, 6)$ and $B = (5, 12)$.

i. Determine the position vector $\boldsymbol{BA}$.

ii. Find the length of the position vector $\boldsymbol{BA}$.

iii. Describe the relationship between the position vectors $\boldsymbol{AB}$ and $\boldsymbol{BA}$.

[5 marks]

b) The two vectors $V1$ and $V2$ of magnitudes $15\ N$ and $20\ N$, respectively. The angle between these vectors is 120 degrees. Calculate the direction and the magnitude of the resultant force using the Parallelogram Law of Vector Addition. [8 marks]

c) If the position vectors of three vectors $\boldsymbol{a}, \boldsymbol{b}$ and $\boldsymbol{c}$ are given as, $\boldsymbol{a} = 2\boldsymbol{i} - \boldsymbol{j} + \boldsymbol{k}$, $\boldsymbol{b} = \boldsymbol{i} + \boldsymbol{j} - 2\boldsymbol{k}$ and $\boldsymbol{c} = \boldsymbol{i} + 3\boldsymbol{j} - \boldsymbol{k}$, then , find λ such that $\boldsymbol{a}$ is perpendicular to the vector $\lambda\boldsymbol{b} + \boldsymbol{c}$. [8 marks]

d) Consider the following vectors in 3-dimensional space:

$$\boldsymbol{a} = (1,2,3)\,, \boldsymbol{b} = (4,5,6)\,, \boldsymbol{c} = (7,8,9).$$

Find the triple product of these 3 vectors $\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c})$. [4 marks]

e) Consider triangle PQR with vertices $P(1,1), Q(4,5)$, and $R(7,2)$. Find the area of the triangle. [5 marks]

[Total – 30 marks]

Question 3

(a) Briefly discuss the difference between linear and non-linear equations. [2 marks]

(b) Consider the function, $f(x) = x^2 - 3$.

i. Check whether there is a root for this function $f(x)$ in the interval $[1,2]$. [4 marks]

ii. Find the number of iterations that should be used in **Bisection Method** to obtain the root for given the tolerance of 0.01. [4 marks]

(c) Find the root of the function $f(x) = x^2$, using **Newton Raphson Method** for **seven** iterations. Here it is given that $x_0 = 4$ be the approximate root of $f(x) = 0$. [10 marks]

[Total – 20 marks]

Question 4

a) Write down the order and the degree of the following differential equations.

i. $\frac{d^3y}{dx^3} + \left(\frac{dy}{dx}\right)^2 = x^2$ [3 marks]

ii. $\left(\frac{d^2y}{dx^2}\right)^3 + \frac{dy}{dx} = e^x$ [3 marks]

b) Find the solution of the ordinary differential equation (ODE):

$\frac{dx}{dt} = 5x - 3$ for $x(t)$. Here it is given the initial condition as $x(2) = 1$.

[14 marks]

[Total – 20 marks]

END OF QUESTION PAPER